

Bhushan Raju Golani

Software Engineer (2+ years) @Amazon → MS Data Science @UMich → Software Intern @Rivian Volkswagen Tech

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📍 Ann Arbor

EDUCATION

University Of Michigan, Ann Arbor

Aug 2025 – May 2027

Master Of Science In Data Science

GPA: 3.96/4

Coursework: Distributed Systems, Advanced Database Management Systems, Probability and Statistics, NLP

Indian Institute of Technology (IIT), Kharagpur

Jun 2018 - May 2023

Bachelor and Master of Technology in Mechanical Engineering (Minor in Computer Science)

CGPA: 8.95/10

Coursework: Algorithms, Operating systems, Object-Oriented programming, Machine Learning, Algorithms Lab (C++)

EXPERIENCE

Graduate Software Engineer Intern | Rivian Volkswagen Technologies | Palo Alto, California May 2026 - Aug 2026

Created a generative-UI pipeline turning agent answers into structured, actionable UI instead of raw chat text.

- Extended the **http-bridge** microservice's single SSE endpoint with an **outputMode** parameter, letting technicians toggle between **2** modes (genUI/text) on **1** endpoint with **zero** breaking changes to existing text clients.
- Opened an **ACP** session between **http-bridge** and a **LangGraph ACP server**, issuing **session/prompt** calls that averaged **7 tool calls per turn** against upstream services, streamed back to the bridge as **session/update** events.
- Buffered answer segment from the agent and forwarded it to a second LLM (**Claude Sonnet**) for markdown-to-**JSONL** UI-spec transformation, adding under **1s** of incremental latency per segment, streaming genUI patches back.
- Persisted each turn's structured UI spec (avg **3,000** characters) to **Aurora PostgreSQL**, delivering a single-stop actionable answer that cut technician triage time by **50%+**.

Agentic AI Researcher | University Of Michigan | Professor Scott Pletcher

Aug 2025 - May 2026

Devised a modular multi-agent platform to automate scientific research and synthesis.

- Engineered a **Research** team of **22** capabilities automating **PubMed**/web retrieval, dedup and relevance ranking
- Designed **VectorDB** team of **21** capabilities with **ChromaDB** semantic/keyword search, clustering, and corpus-resolution
- Established **Writing** team of **12** capabilities producing grants and manuscripts through review/revise loops.
- Crafted a **master agent** decomposing each query into per-team **execution plans**, orchestrating **6** specialized teams and passing typed evidence packs losslessly across boundaries via **shared artifacts**.

Software Engineer | Amazon | Alexa Sensitive Content Intelligence Team

Jul 2023 - Jul 2025

Built an end-to-end event-driven pipeline to ingest, enrich, classify, and triage Alexa user feedback at scale.

- Developed scalable cloud services in **Java** and **AWS**, achieving **99.99%** uptime while processing **50,000+ TPS**.
- Created an ingestion service extracting **10+** fields into DynamoDB, publishing structured events via **SNS** (pub/sub).
- Built a downstream microservice that subscribes via **message queue (SQS)** to feedback ingestion SNS & integrates with a dialog history fetcher service via **API** to enrich feedback with context (3 turns), persisting it to a new DynamoDB.
- Engineered a classification handler that uses **Claude Sonnet 3.5** via **Amazon Bedrock** to perform prompt-based classification (e.g., sentiment, policy violations) of user feedback, storing results to DynamoDB for further use.
- Designed the Risk Handler (**HLD + LLD**), integrating with embedding generator service via API to retrieve vector representation of text and finding risk score by combining frequency and LLM-derived severity with **<2s** latency.
- Integrated a service that reads through all the above-mentioned 4 databases and indexes them into **ElasticSearch**.

Software Engineer Intern | Amazon | In-Ads-Tech, Team

May 2022 - Jul 2022

Implemented an automated serverless pipeline to fetch, filter, and credit Deutsche Bank remittance balances.

- Configured an **Event Driven System (EventBridge)** to trigger an API call every 4 hours to retrieve remittance files.
- Devised a microservice to extract tax details and advertiser info via internal **authorized & authenticated API calls**.
- Worked on a post-processing Lambda to update advertiser records with extracted data for remittance tracking.
- Achieved **less than 1s latency** and deployed production-ready code with **99% uptime**.

TECHNICAL SKILLS

Languages - Java, C++, C, SQL, JavaScript, TypeScript, Python.

Technologies - ReactJS, ElasticSearch, AWS (EC2, Lambda, S3 etc), REST APIs, Git, Docker, CI/CD, Golang

PUBLICATIONS

Mapping fluid structuration to flow enhancement in nanofluidic channels

Special Collection: Machine Learning Hits Molecular Simulations | The Journal of Chemical Physics

- Designed physics-informed neural network models to predict flow enhancement factors in nanofluidic channels.

Prediction of Cloud Server Job Failures using Machine Learning based KNN Classification and LSTM Modelling

International Journal of Engineering Research and Technology (IJERT) | TATA Communications Internship

- Developed a KNN classification and LSTM modeling based pipeline to anticipate cloud job failures with **94%** accuracy.